

Ashish Kumar Jha

Backend-Focused Software Engineer | Java | Spring Boot | Distributed Systems
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Summary

Backend-focused Software Engineer building Java/Spring Boot microservices and distributed systems with Kafka, Redis, MySQL, concurrency control, and event-driven architectures. Hands-on AI/ML experience spanning LLM fine-tuning, RAG, vector search, and Python-based model workflows.

Education

Geeta University

B.Tech in Computer Science & Engineering

2022–2026

Panipat, Haryana

Experience

FixFin Technologies

Software Development Intern

June 2024–August 2024

Siliguri, West Bengal

- Optimized complex SQL queries for operations-reporting workflows, reducing data-retrieval latency by 40%.
- Integrated backend REST APIs across frontend modules and built responsive interfaces using React, HTML, CSS, and JavaScript.

Projects

E-Wallet Backend

Repository

- Built a Java 21/Spring Boot microservices wallet platform with API Gateway, User, Wallet, Transaction, and Notification services backed by isolated MySQL databases.
- Prevented duplicate transfers and concurrent balance races using idempotency keys and `SELECT FOR UPDATE` pessimistic locking; used synchronous REST for transfer execution and Kafka for asynchronous notifications.
- Added Redis balance/user caching, Resilience4j circuit breaking, Dockerized deployment, GitHub Actions CI/CD, and AWS EC2 hosting.

Flash Sale Engine

Repository

- Engineered a high-concurrency Java 21/Spring Boot purchase system using Redis atomic inventory controls, MySQL transactions, JWT security, and Kafka-based asynchronous order processing.
- Prevented overselling under parallel requests and added per-user Redis rate limiting; validated the purchase flow with 1,000+ JMeter threads for stability and response-time testing.

Codebase Q&A – RAG System

Repository

- Built four Spring Boot services for repository ingestion, embedding, retrieval, and gateway routing; asynchronous Kafka processing decouples code ingestion from vector generation.
- Implemented semantic code search with Qdrant embeddings and Redis response caching, retrieving top-k relevant code chunks to generate contextual natural-language answers.

TinyLlama AI/ML Tutor

Repository

- Fine-tuned TinyLlama-1.1B-Chat-v1.0 with 4-bit NF4 QLoRA and supervised fine-tuning using Transformers, PEFT, TRL, and PyTorch on 2,000 synthetic AI/ML tutoring examples.
- Built the end-to-end data, training, inference, and evaluation pipeline; 200 held-out test examples yielded 0.625 evaluation loss and 0.820 mean token accuracy, with a FastAPI comparison service.

Technical Skills

Languages: Java, Python, C++, JavaScript, SQL

Backend & Frameworks: Spring Boot, Spring Security, Microservices, REST APIs, JPA/Hibernate, Resilience4j, System Design

Databases & Messaging: MySQL, Redis, Apache Kafka

AI/ML & LLM: PyTorch, Hugging Face Transformers, PEFT, TRL, QLoRA, LoRA, SFT, RAG, Embeddings, Qdrant

Tools & Cloud: Git, GitHub Actions, Docker, Docker Compose, AWS EC2, Maven, Swagger/OpenAPI, Apache JMeter

Competitive Programming

LeetCode — 250 solved, Max Rating 1654

GeeksforGeeks — 300+ solved

Codeforces — Pupil, Max Rating 1381

CodeChef — 2-Star, Max Rating 1514

AtCoder — 451, 8 Kyu